

Cryptography (CS402/MA492)

What is this?

- Cryptography is the "science of secret writing"

kryptós: hidden, secret graphem: to write

Goal: hide the meaning of a message

- Cryptanalysis is the science (and art!) of "breaking"

anályein: to loosen, to untie cryptographic methods

Goal: gain access to or recover secret messages without prior knowledge of the "key" used to produce them

(This need not be dodgy! To prove that a cryptographic method is "secure", we should try to break it.)

- Cryptology consists of cryptography & cryptanalysis.

logia: study

(Q: What do the terms underlined in red really mean.)

What will we do?

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We'll consider both cryptography and cryptanalysis; most lectures will focus on the former.

Topics	Mathematical techniques
"historical cryptography"	elementary / ad hoc
symmetric ciphers	basic probability theory, polynomials and finite fields
asymmetric ciphers / public-key cryptography	modern (computational) number theory (e.g. primality testing, elliptic curves, ...)

Slogan: this is "applied pure mathematics"

Prerequisites?

We almost start from scratch. The following might be useful though:

- Matrices, divisibility, congruences & modular arithmetic.
- Very basic computer programming... I'll use Python 3.

Literature

There are numerous good texts available. The following are all suited as companions for this module:

- Paar, Petzl: Understanding Cryptography
- Smart: Cryptography, An Introduction
- Stinson, Paterson: Cryptography: Theory and Practice.

Mathematically-minded people should also read

- Koblitz: A Course in Number Theory and Cryptography.

Note: Further references on the module web page!

Practical matters

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Course web page: <http://www.maths.nuigalway.ie/~rossmann/cs402>

Lecturer: Tobias Rossmann Tobias.rossmann@nuigalway.ie

Office: ADB-1006

Office hours: Wed 1-2, Thu 2-3

Tutorials: Tue 3-4

Venues: TBA

Wed 2-3

Every other week, beginning in week 3.

Tutorials will provide further examples, illustrate key ideas, etc.

Assessment: 70% written exam, 3 × 10% online assignments

Assignments: Focus on the practical side, computer use will be essential. Complements lectures.

The Caesar cipher

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"If he had anything confidential to say, he wrote it in cipher, that is, by so changing the order of the letters of the alphabet, that not a word could be made out.

If anyone wishes to decipher these, and get at their meaning, he must substitute the fourth letter of the alphabet, namely D, for A, and so with the others."

— Suetonius, "The Twelve Caesars" (AD 121)

Table:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Ex: CRYPTOGRAPHY
FUBSWRJUDSKB

Security issue: If an adversary knows the method used, (s)he can decode all messages.

Shift cipher: Replace the cyclic shift by 3 by some other number ... 26 choices ("Keys"), one useless (the "0-shift").

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Goal: The central objective of cryptography is to enable two parties (usually called Alice and Bob) to communicate over an insecure channel (e.g. telephone, internet) in a way that prevents a third party (an eavesdropper Eve, an opponent Oscar) from understanding what is being said.

Other major goals: Ensure the integrity and authenticity of messages.

Has the message been altered?

Was it really Alice who sent the message?

Getting formal:

Defn A cryptosystem or cipher consists of

- a set P of plaintexts,
- a set C of ciphertexts,
- a set $\mathcal{K}$ of keys (the key space), and

for each key $K \in \mathcal{K}$, functions $e_K: P \rightarrow C$ (encryption or encrypting function) and $d_K: C \rightarrow P$ (decryption/decrypting) such that $d_K(e_K(x)) = x$ for all $x \in P$.